

Session 5 · Thursday 10 September 2026 · Two in the room, two on the problem set. Start wherever the scaffolding stops helping you.

Getting a removal rate out of decay data

Four versions of one procedure, with **less of my working shown each time**. We do items 1 and 2 in the room — eight minutes, and item 1 is already done. Items 3 and 4 are on PS2; they are there rather than here because each needs a decision defended in prose, and this room is worth more than that.

Both tables below are the same experiment: at $t = 0$ production is switched off, and what is already in the cell is removed. The cells **keep growing** throughout. That sentence is the whole of item 2.

HOST	$T_d = 30$ min, so $\mu = \ln 2/30 = 0.0231$ min ⁻¹	SIGNAL	y = fluorescence, arbitrary units
MODEL	$dp/dt = -(\gamma + \mu)p$ once production stops	SO	$\ln y$ is linear in t with slope $-(\gamma + \mu)$

These are the synthetic traces from `posb.data.decay_timecourse`, thinned to seven points so you can do this on paper. PS2 gives you all forty and asks you to fit them in code.

1 • Fully worked — the LAA tag

t (MIN)	0	10	20	30	40	50	60
y	101.5	68.0	44.9	29.5	20.3	13.5	8.8
$\ln y$	4.62	4.22	3.80	3.38	3.01	2.60	2.17

Take logs, and check they fall on a line. The third row is already there. The differences between consecutive entries are $-0.40, -0.42, -0.42, -0.37, -0.41, -0.43$ — constant to within the scatter, which is what "exponential" means and is the only evidence you have that the model is the right one.

Read the slope. Over the whole run, $\Delta \ln y = 2.17 - 4.62 = -2.45$ across $\Delta t = 60$ min:

$$\text{slope} = \frac{-2.45}{60} = -0.0408 \text{ min}^{-1}$$

Name what you just measured. The slope is $-(\gamma + \mu)$, so the **removal rate constant** is $k = 0.0408 \text{ min}^{-1}$. It is a rate constant. Its units are per minute.

Convert to a half-life — and this is where the marks go.

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{0.0408} = 17.0 \text{ min}$$

Not 25 minutes. $1/k = 24.5 \text{ min}$ is the *e*-folding time, a different number for a different question. The factor of **$\ln 2$** is the single most common dropped mark in this course, and units will not catch it for you — both answers come out in minutes.

► **Why does that step follow?** You fitted a straight line to **$\ln y$** , not to **y** . Give the reason in terms of the *noise*, not the algebra: what is being assumed about the size of the measurement error, and why does a fluorescence reading satisfy it?

2 • The last steps are yours — the AAV tag

Same host, same experiment, same growing cells.

t (MIN)	0	10	20	30	40	50	60
y	100.4	68.0	51.7	35.4	24.4	17.3	12.4

Take logs, read the slope, and report k and $t_{1/2}$, with units on both.

$\ln y$:

Now the question the number does not answer. You have one exponential and one rate constant. Andersen et al. report the AAV tag as a **60 minute** half-life. Your $t_{1/2}$ is not 60 minutes and is not supposed to be.

► **Is what you fitted γ , or $\gamma + \mu$?** Answer from the *description of the experiment*, not from the number — say which sentence at the top of this page decides it.

► **Now recover Andersen's number.** Using $\mu = 0.0231 \text{ min}^{-1}$, extract γ from your k and convert it to a degradation half-life. You should land close to 60 min. Write the two lines of arithmetic.

► **And the transferable habit.** Andersen measured his half-lives after a **medium downshift**, which arrests growth. Say in one sentence why that makes his numbers safe to add to a μ of *your* choosing, and why a half-life measured in growing cells could not be reused that way.

On the problem set — the same procedure, unscaffolded

These are **PS2 Q2 and Q5**. Each is the place where the number you just extracted has to be used for something.

3 • The design question — PS2 Q2

You need a reporter with a response time of $t_{1/2} = 12$ min, in a host that divides every 25 minutes. Which Andersen tag do you choose?

State what you are solving. Write down the γ that the specification demands, then convert it to the degradation half-life a tag would have to have.

Now you: everything else. Andersen's four tags are ASV 110, AAV 60, LVA 40, LAA 40 minutes. Decide whether the specification can be met, and if it cannot, report what the **best** available tag actually gives you — both the response time and the steady-state level relative to untagged.

The interesting half of this question is the part after "no". A specification you cannot meet by choosing a part is not a dead end; it is a question about which variable you were treating as fixed. Two of them are not.

4 • Bare problem — PS2 Q5; BioE 247. 147 may attempt it for extra credit

Everything above assumed production switches on as a step. It does not: inducer has to enter the cell and the promoter responds over some time. Let production ramp instead —

$$\frac{dp}{dt} = \alpha \left(1 - e^{-t/\tau_{\text{ind}}}\right) > -(\gamma + \mu)p, \quad p(0) = 0$$

Solve for $p(t)$, show it reduces to the step response as $\tau_{\text{ind}} \rightarrow 0$, and say whether the response time gets longer or shorter.

Then the part that is actually being asked: **give the condition under which today's formula $t_{1/2} = \ln 2 / (\gamma + \mu)$ is still a good description**, stated as a comparison between two timescales in the style of session 4.

A slope is a rate. A rate is not a half-life. And a half-life quoted without saying whether the cells were growing is a number you cannot move to another host — which means most of the ones you will read are.